TECHSHIP INTERNSHIP

Day 3- Functions,Arrays,Objects & ES6 Concepts

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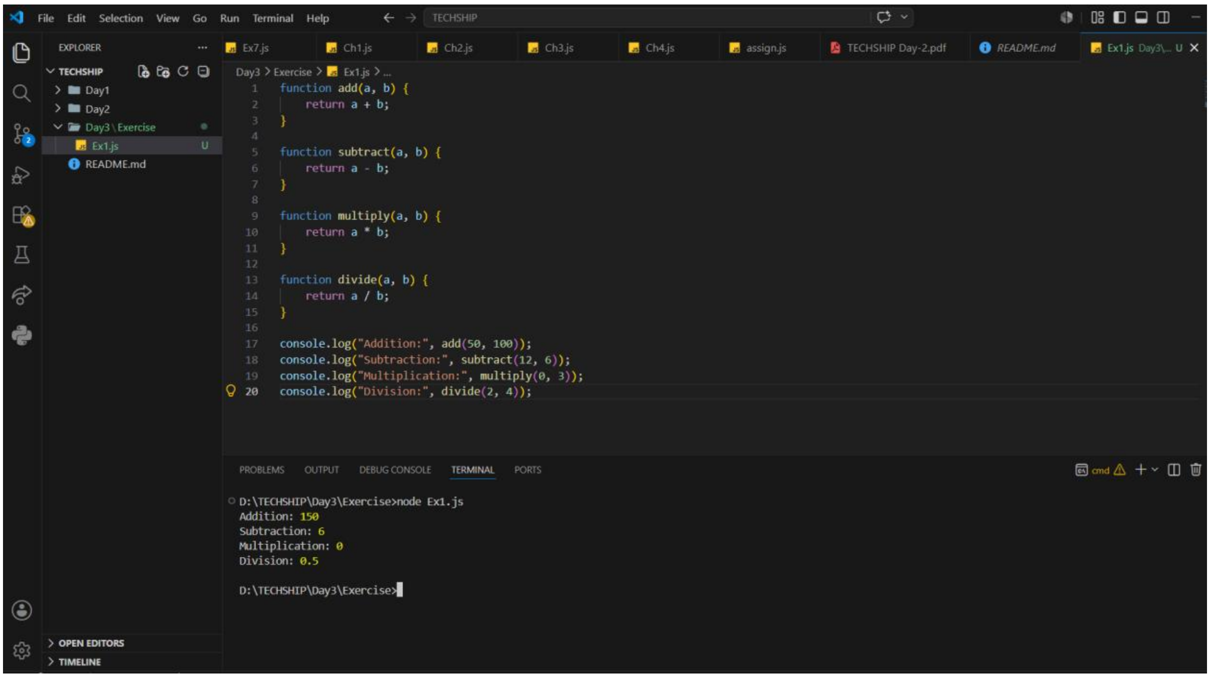
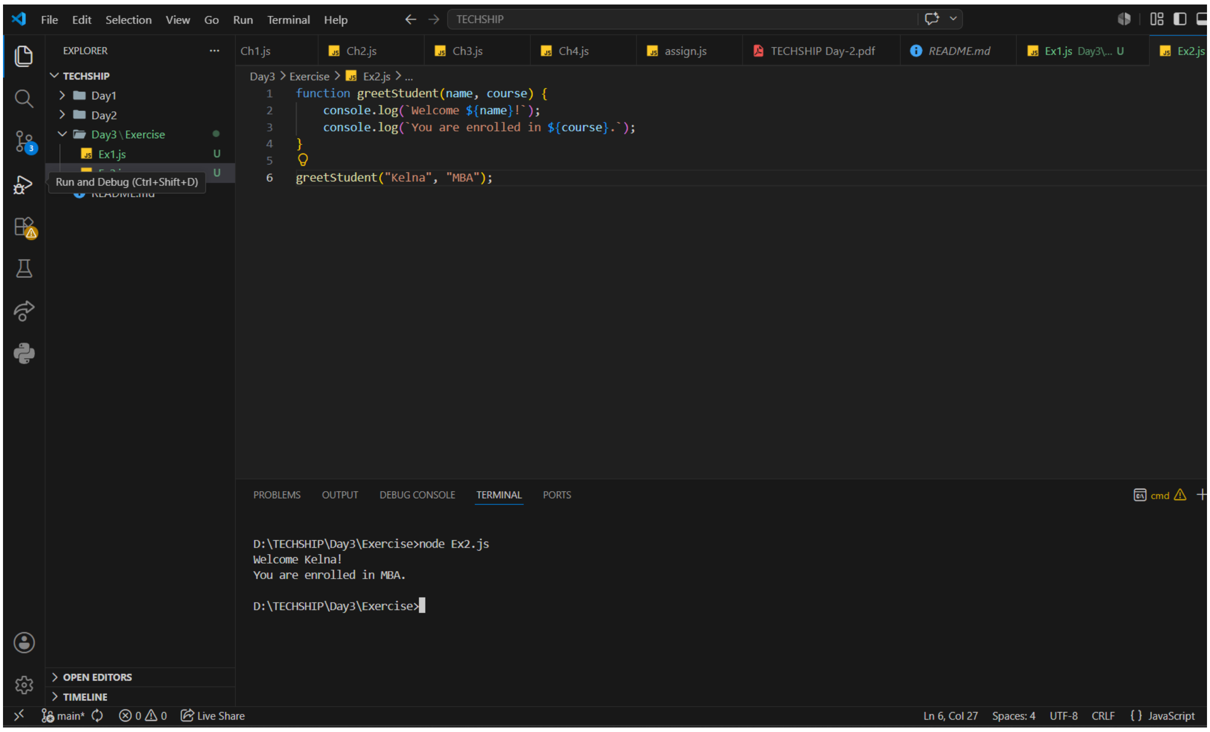
During Day 3 of the internship, I learned how to work with functions, arrays, objects, and modern JavaScript ES6 features. I gained a better understanding of how reusable functions help organize code and how arrays and objects can be used to store and manage structured data efficiently. I also explored array methods such as forEach(), filter(), map(), and reduce(), which are useful for processing and manipulating data in real-world applications.

The ES6 feature I found most useful was the spread operator because it allows objects and arrays to be copied and modified easily while keeping the original data unchanged. I also found template literals very helpful for generating clean and readable reports.

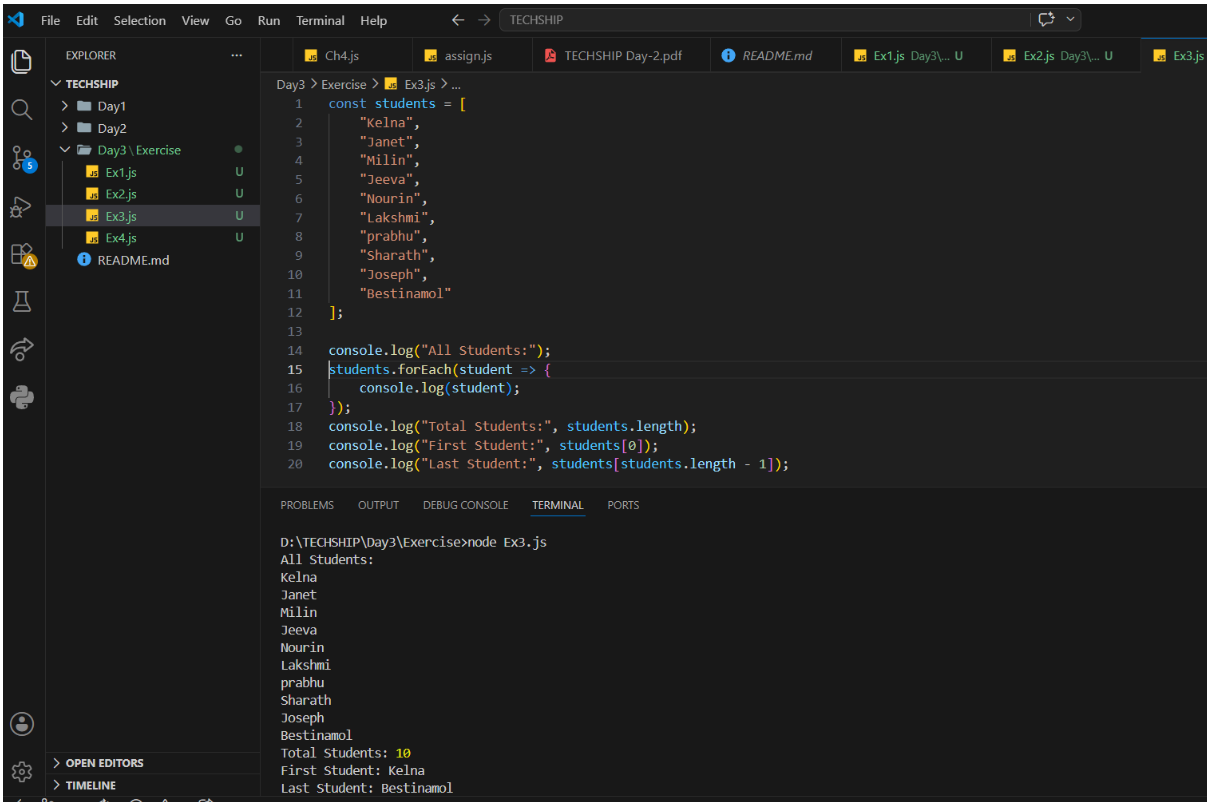
The main challenge I faced was understanding the reduce() method, particularly how its parameters work and the purpose of the initial value. While implementing the Student Management System assignment, I initially found it difficult to calculate averages and identify the top performer using array methods. However, after practicing with examples and debugging the code, I became more confident in using these concepts. Overall, this session improved my understanding of structured JavaScript programming, strengthened my problem-solving skills, and gave me hands-on experience in developing programs that resemble real-world data management systems.



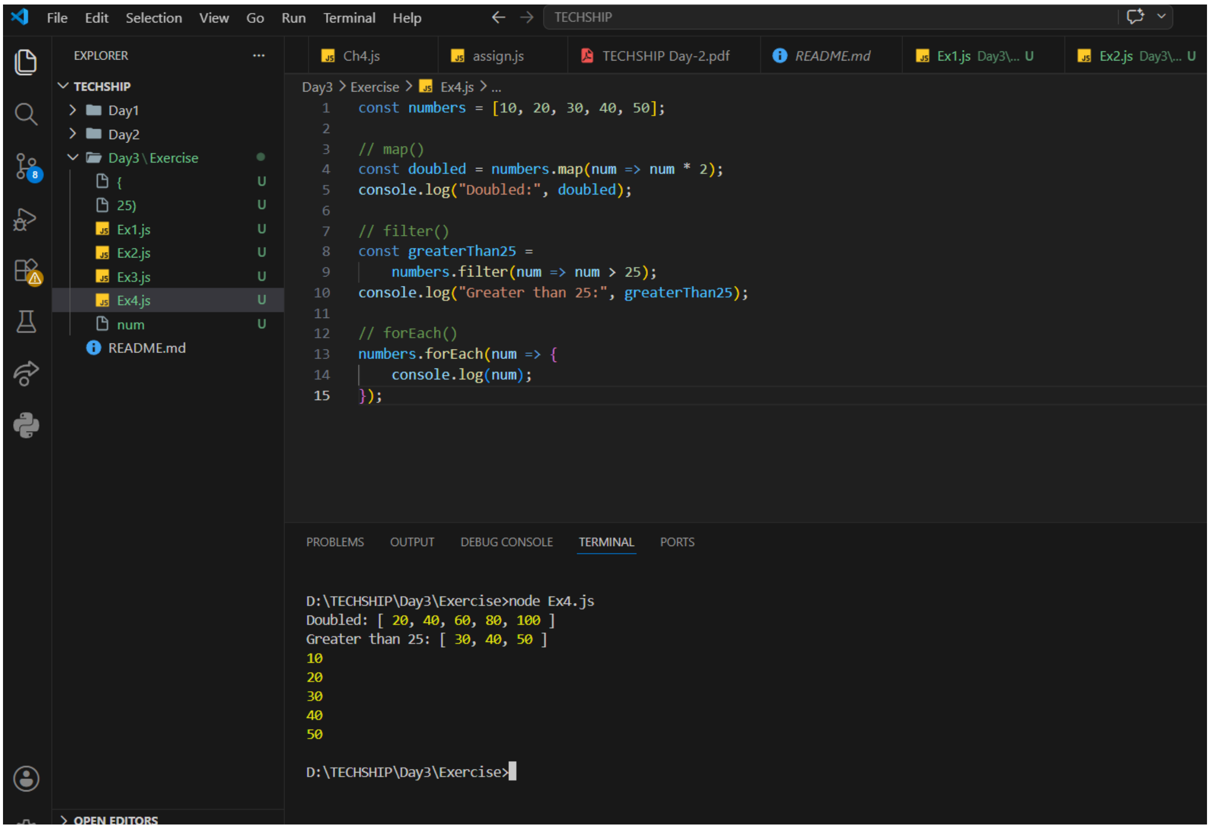
Ex1:Function Based Calculator



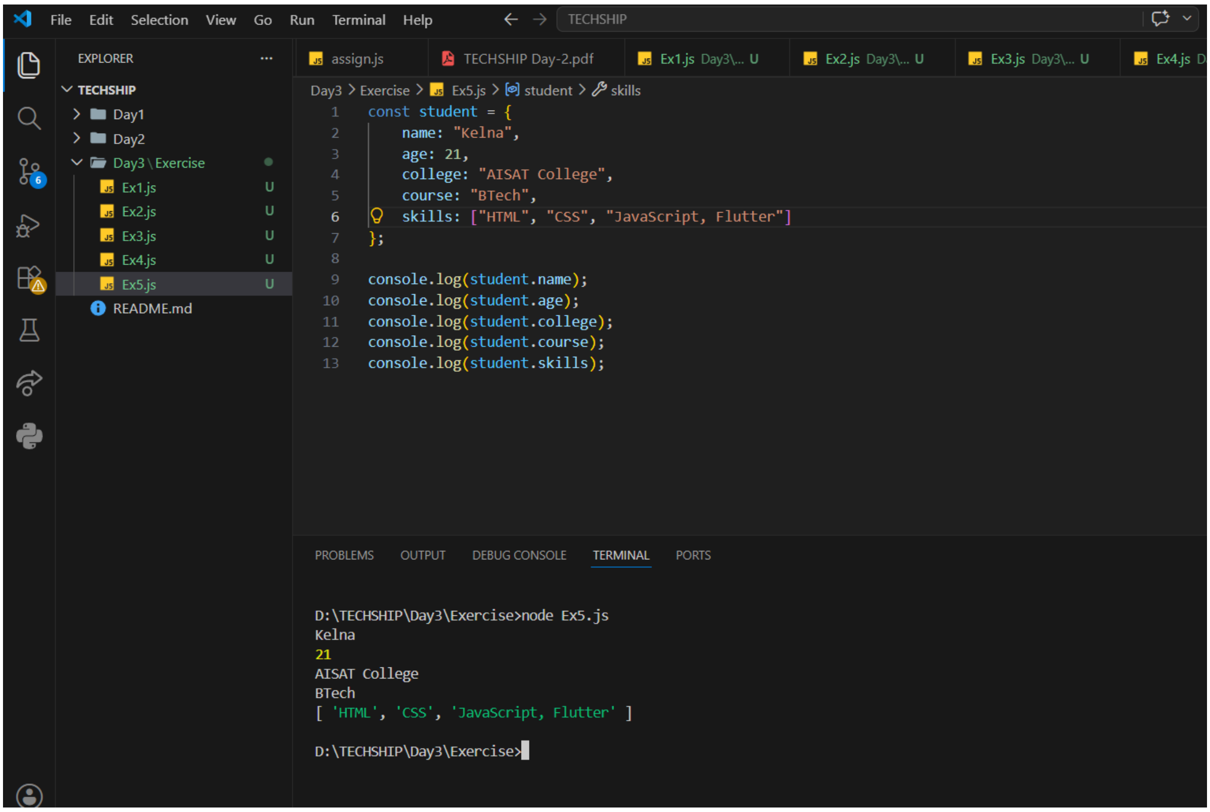
Ex2:Student Greeting Function



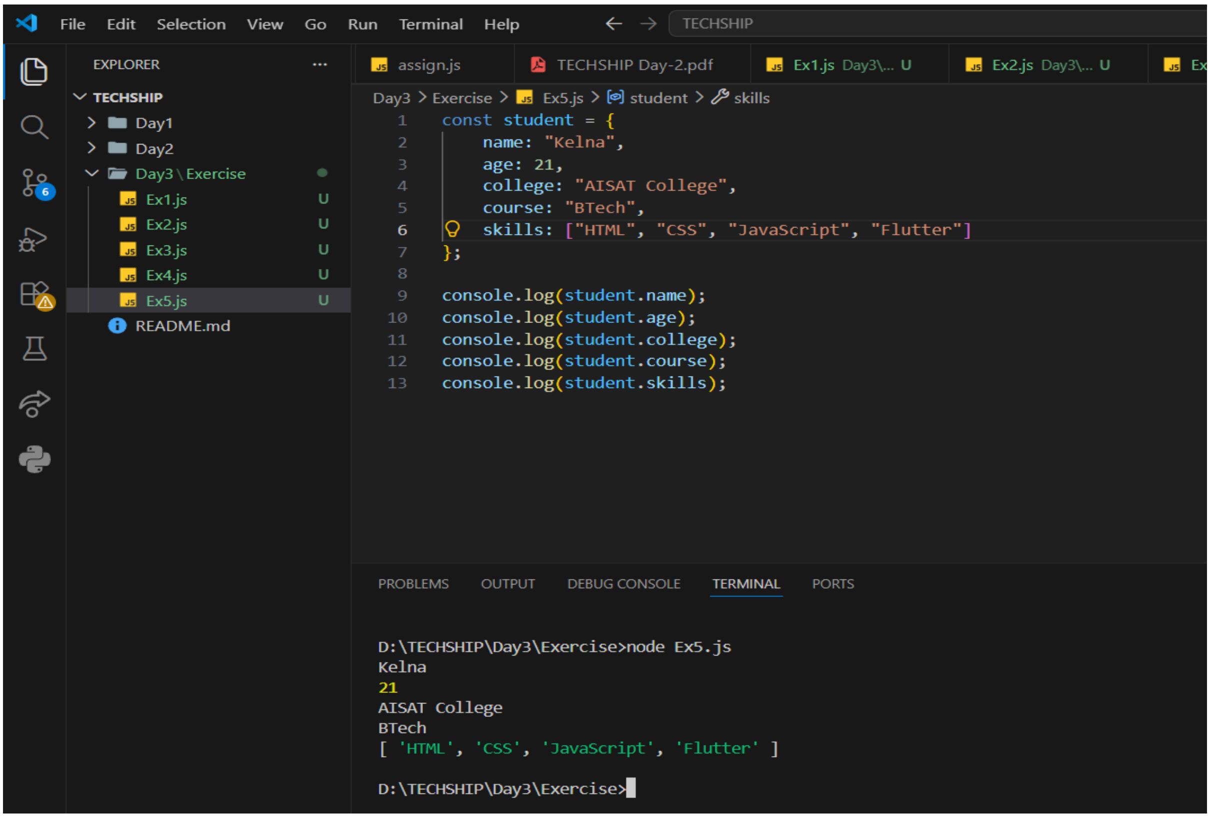
Ex3:Working with Arrays



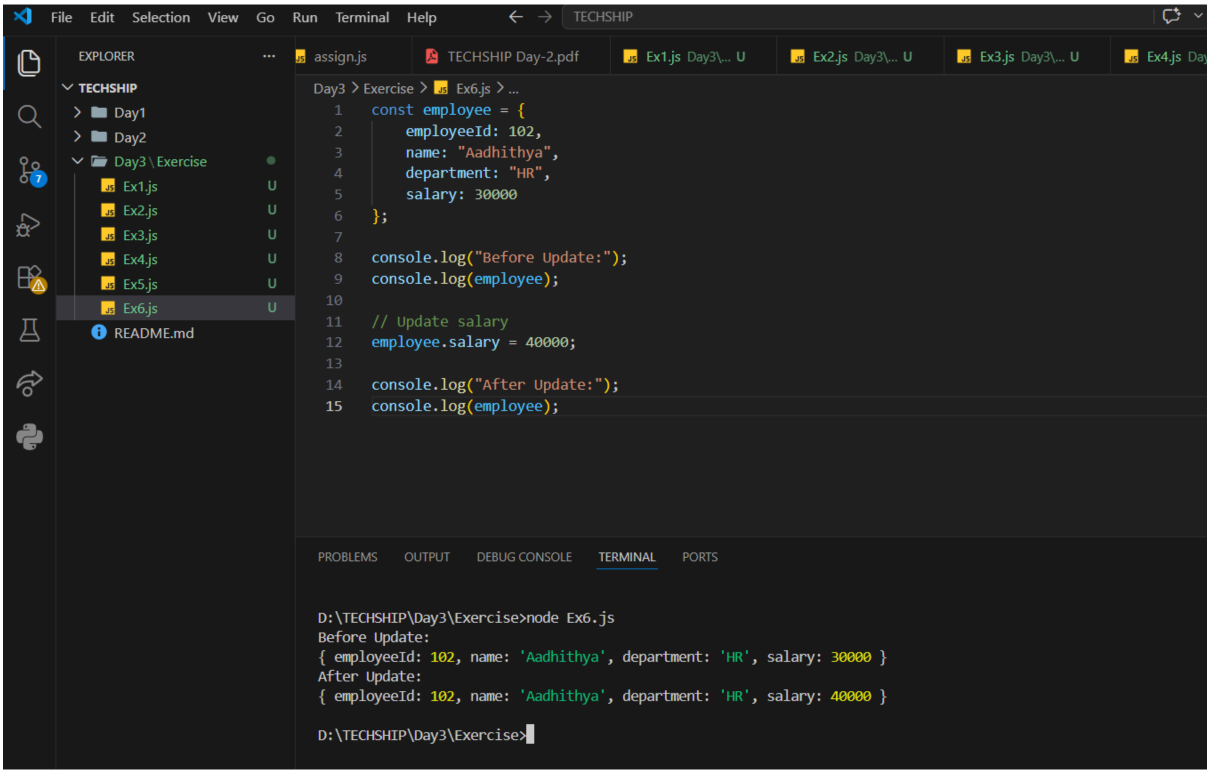
Ex4:Array Methods



Ex5:Student Object

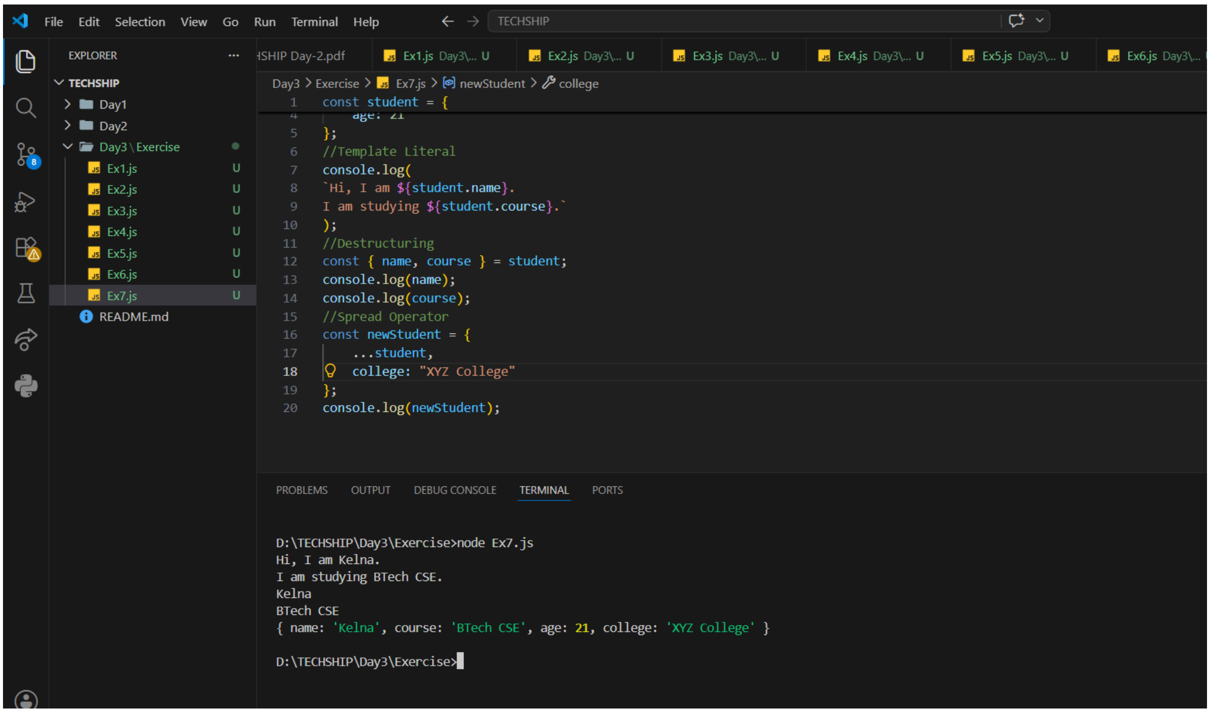


Ex6:Employee Object

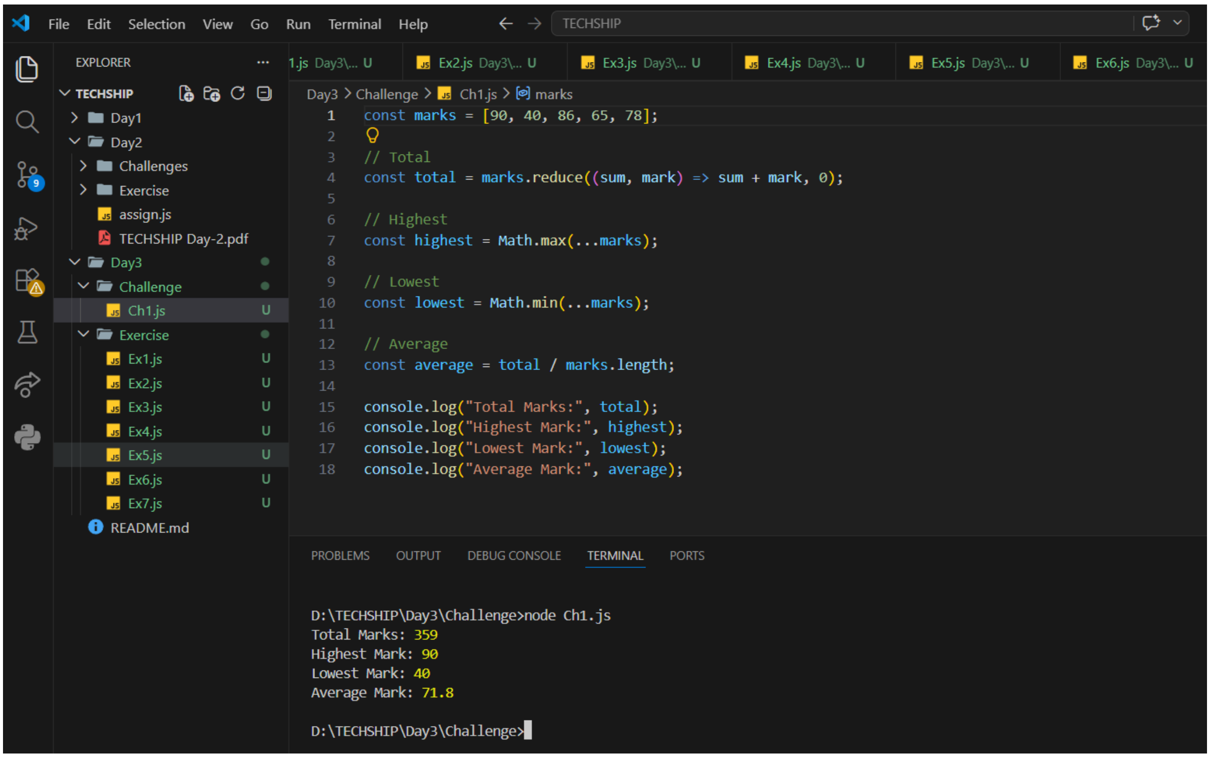


Ex7:ES6 Practise

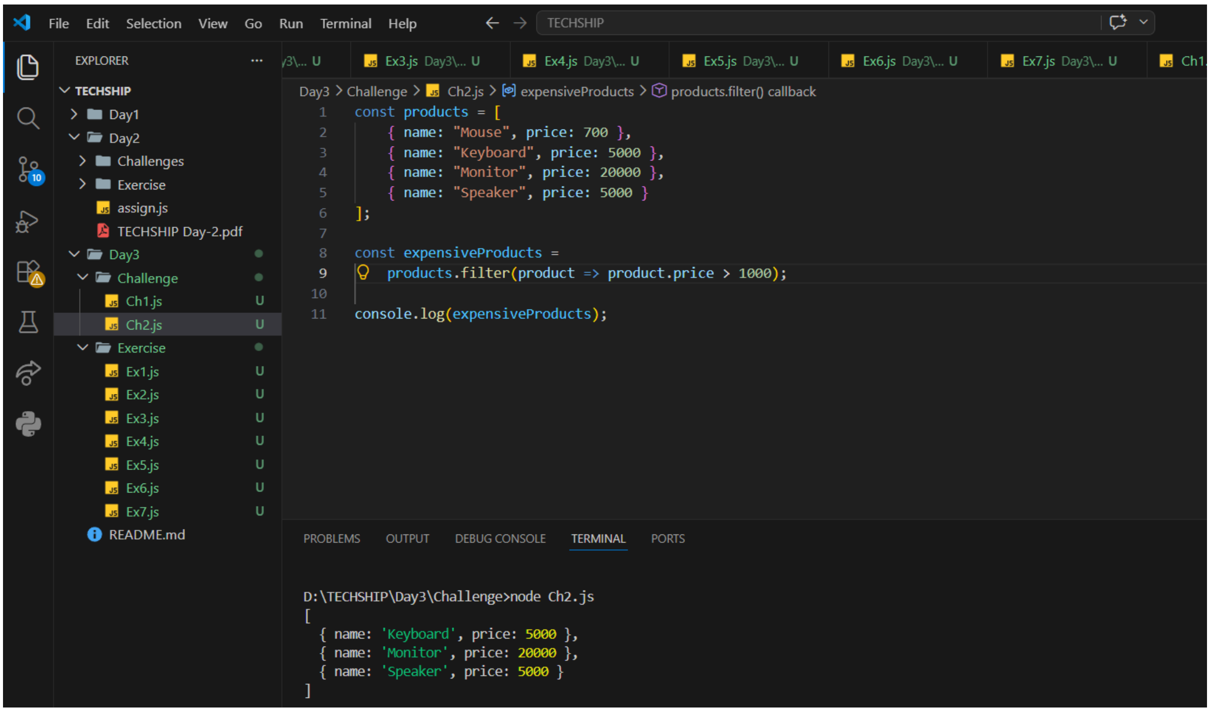
Challenge Problems



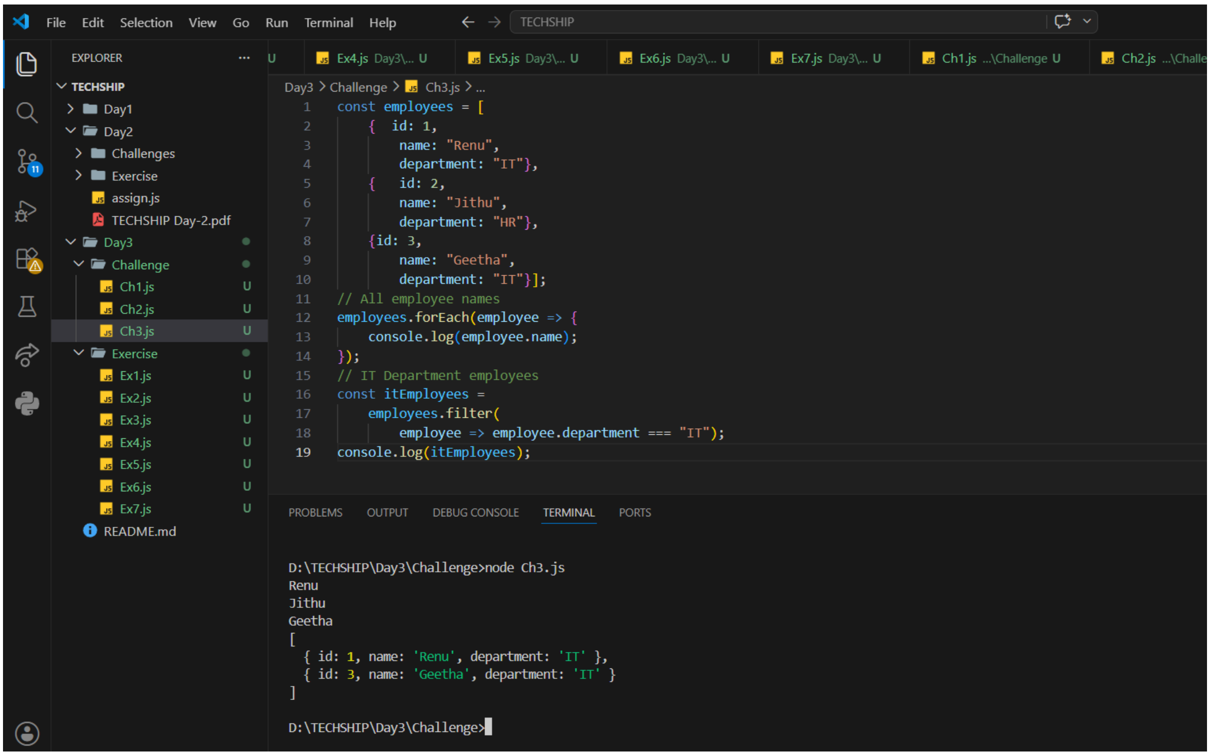
1.Student Marks Analyser



2.Product Filter System



3.Employee Directory



4.Course Enrollment System

Assignment

Ex6.js Day3\... U | Ex7.js Day3\... U | Ch1.js ...\Challenge U | Ch2.js ...\Challenge U | Ch3.js ...\Challenge U | Ch4.js ...\Challenge U

Day3 > assign.js > studentAverages > students.map() callback

```javascript
const students = [{id: 100,
        name: "Evelin",
        age: 20,
        course: "Cardiology Tecbnician",
        marks: [75, 80, 70, 85, 90]},{
        id: 101,
        name: "Ann",
        age: 21,
        course: "Btech EC",
        marks: [79, 82, 75, 88, 80]},{
        id: 102,
        name: "Janet",
        age: 21,
        course: "BTech CS",
        marks: [90, 95, 92, 88, 91]},{
        id: 103,
        name: "Kelna",
        age: 22,
        course: "BTech AI/ML",
        marks: [65, 70, 68, 72, 66]},
    { id: 104,
        name: "Divya",
        age: 21,
        course: "Bsc Nurse",
        marks: [78, 80, 75, 82, 79]}];
        // Calculate average for each student
const studentAverages = students.map(student => {
const total = student.marks.reduce(
        (sum, mark) => sum + mark,
        0);
const average = total / student.marks.length;
return {
        ...student,
        average: average};});
// Students scoring above 75%
const topStudents = studentAverages.filter(
    student => student.average > 75);
// Find top performer
const topper = studentAverages.reduce(
    (highest, student) =>
        student.average > highest.average
            ? student
            : highest);
console.log("================================");
console.log("STUDENT MANAGEMENT REPORT");
console.log("================================");
// Display all students with report
studentAverages.forEach(student => {let status;
if (student.average >= 80) {
        status = "Excellent";
    } else if (student.average >= 70) {
        status = "Good";
    } else {
        status = "Needs Improvement";}
console.log(`ID: ${student.id}
Name: ${student.name}
Course: ${student.course}
Average Marks: ${student.average.toFixed(2)}
Status: ${status}`);});
// Students above 75%
console.log("--------------------------------");
console.log("Students Scoring Above 75%:");
topStudents.forEach(student => {console.log(`${student.name} - ${student.average.toFixed(2)}`);});
// Top performer
console.log("--------------------------------");
console.log(`Top Performer: ${topper.name}`);
console.log(`Highest Average: ${topper.average.toFixed(2)}`);
console.log("================================");
```

Student Management System